

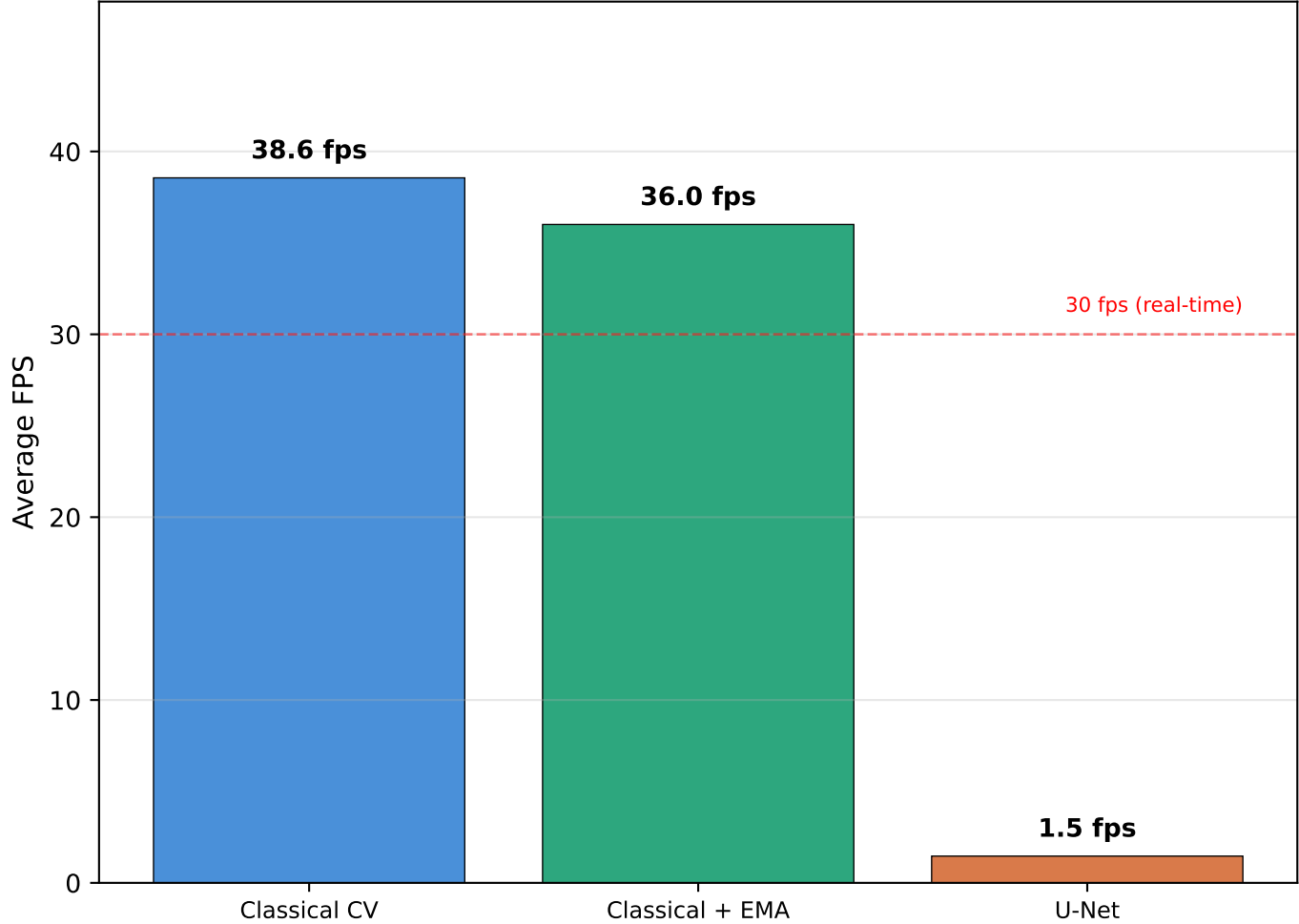
(A) TuSimple Benchmark — Detection Quality

Lane Detection Method Comparison on TuSimple Benchmark

Method	Accuracy (SOTA convention)	Matched-lane precision	FPR	FNR
Classical CV (ours)	25.13%	98.99%	48.11%	73.91%
U-Net (ours)	39.76%	99.53%	20.94%	58.97%
UFLD [Qin 2020]	96.06%	—	~5%	~5%
SCNN [Pan 2018]	96.53%	—	~6%	~6%

Evaluated on 100 test frames. Green = better between our methods. SOTA accuracy is comparable to published numbers.

(B) Throughput on CPU (100 frames)



(C) Temporal Smoothing reduces jitter (4.1× average reduction)

